Example 9.19

Evaluate $\int_0^{\frac{\pi}{4}} \frac{1}{\sin x + \cos x}\, dx$

Solution

$$I = \int_0^{\frac{\pi}{4}} \frac{1}{\sin x + \cos x}\, dx = \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right)}\, dx$$

$$= \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \frac{1}{\left(\cos\frac{\pi}{4}\cos x + \sin\frac{\pi}{4}\sin x\right)}\, dx = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \frac{1}{\cos\left(\frac{\pi}{4} - x\right)}\, dx$$

$$= \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \frac{1}{\cos x}\, dx \text{ , since } \int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$= \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \sec x\, dx = \frac{1}{\sqrt{2}} \left[\log(\sec x + \tan x)\right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{\sqrt{2}} \left[\log(\sqrt{2}+1) - \log(1+0)\right]$$

$$= \frac{1}{\sqrt{2}} \log(\sqrt{2}+1).$$

Property 7

$$\int_0^{2a} f(x)\, dx = \int_0^a \left[f(x) + f(2a-x)\right] dx.$$

Proof

By property 3, we have $\int_0^{2a} f(x)\, dx = \int_0^a f(x)\, dx + \int_a^{2a} f(x)\, dx$. (1)

Let us make the substitution $x = 2a - u$ in $\int_a^{2a} f(x)dx$. Then, $dx = -du$.

When $x = a$, we have $u = 2a - a = a$. When $x = 2a$, we have $u = 2a - 2a = 0$. So, we get

$$\int_a^{2a} f(x)dx = \int_a^0 f(2a-u)(-du) = \int_0^a f(2a-u)du = \int_0^a f(2a-x)dx. \quad \ldots(2)$$

Substituting equation (2) in equation (1), we get

$$\int_0^{2a} f(x)\, dx = \int_0^a f(x)\, dx + \int_0^a f(2a-x)\, dx$$

$$= \int_0^a \left[f(x) + f(2a-x)\right] dx.$$

Property 8

If $f(x)$ is an even function, then $\int_{-a}^a f(x)\, dx = 2\int_0^a f(x)\, dx$.

(Recall that a function $f(x)$ is an even function if and only if $f(-x) = f(x)$.)

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